

The impact of intestinal microbiota in antithymocyte globulin-based myeloablative allogeneic hematopoietic cell transplantation

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BACKGROUND: The correlation between intestinal microbiota and clinical outcomes after allogeneic hematopoietic stem cell transplantation (allo-HCT) has been reported in platforms with T-cell depletion or postcyclophosphamide-based graft-vs-host disease (GVHD) prophylaxis regimens. It is still unclear whether it is the same in platforms of antithymocyte globulin (ATG)-based myeloablative allo-HCT. **METHODS:** A total of 603 fecal specimens from 100 consecutive patients receiving allo-HCT were collected between December 2018 and July 2020. Fecal samples were profiled with next-generation sequencing of bacterial 16S ribosomal RNA (rRNA) genes. **RESULTS:** The diversity decreased to the lowest level at approximately day 12 after allo-HCT and then increased over time. According to the diversity of 314 samples that were collected from 86 patients during the engraftment period, patients were grouped into the low- and high-diversity groups. Two-year overall survival in the high-diversity group was significantly longer than that in the low-diversity group (83.7% vs 60.6%, $P = .026$). Further analysis revealed that worse outcomes for patients with low diversity were associated with increased risk of worse outcomes for patients with low diversity (adjusted hazard ratio, 4.95; $P = .046$). Its association with relapse and GVHD was not found. Compositional analysis of fecal microbiota revealed that the abundance of bacteroides decreased greatly during allo-HCT, whereas that of *Enterococcus*, *Klebsiella*, and *Escherichia* was found to be increased. **CONCLUSIONS:** This study indicates that gut dysbiosis in platforms of ATG-based myeloablative allo-HCT featured loss of bacterial diversity. The diversity of the intestinal flora at the engraftment period was an independent predictor of longer survival. **Cancer** 2022;128:1402-1410. © 2022 American Cancer Society.

LAY SUMMARY:

- The correlation between intestinal microbiota and clinical outcomes after allogeneic hematopoietic stem cell transplantation (allo-HCT) is reported in platforms with T-cell depletion or postcyclophosphamide-based graft-vs-host disease (GVHD) prophylaxis regimens.
- It is still unclear whether it is the same pattern in platforms of antithymocyte globulin (ATG)-based T-cell repletion myeloablative allo-HCT.
- Our study indicated that gut dysbiosis in platforms of ATG-based myeloablative allo-HCT also features loss of bacterial diversity. The diversity of the intestinal flora at the engraftment period is an independent predictor of longer survival.

KEYWORDS: allogeneic hematopoietic stem cell transplantation, antithymocyte globulin (ATG), intestinal microbiota, microbiota diversity, T-cell repletion.

INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HCT) remains a curative option for patients with hematologic malignancies. The success of this technique depends on both the antitumor effects of the conditioning regimen and the immune-mediated graft-versus-tumor effects by alloreactive lymphocytes of the grafts. Full use of this procedure is still limited by infection, multiorgan failure, graft-vs-host disease (GVHD), and relapse, especially for older patients.

The human gut microbiota plays an important role in physiology and diseases.¹⁻³ The intestinal microbiota changed dramatically throughout the course of transplantation, which was characterized with profound loss of diversity and expansions of potentially pathogenic bacterial taxa.^{4,5} This alteration is mainly because of the mucosal

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barrier damage associated with the conditioning regimen and the routine administration of various antibiotics in the prolonged neutropenia period.^{6,7} The correlation between the intestinal microbiota during the engraftment period and major outcomes after allo-HCT has been reported.^{5,8-10} However, all these characteristics were mainly in platforms with calcineurin inhibitor-based T-cell-repleted platforms or ex vivo CD34-selected T-cell-depleted platforms, along with a minority using postcyclophosphamide-based T-cell-repleted GVHD prophylaxis regimens. Transplantation procedures including antibiotic use, conditioning regimen, and GVHD prophylaxis vary across transplantation centers.^{5,11,12} In addition, intestinal microbiota was also affected by factors such as geographical environment, dietary habits, and race.^{13,14} To date, there are few studies that featured the pattern of the intestinal microbiota over allo-HCT and its correlation with clinical outcomes in China, where antithymocyte globulin (ATG)-based GVHD prophylaxis is now the mainstream. Currently, the annual number of allo-HCT cases is about 10,000 in China. About one-half of these cases were from human leukocyte antigen-haploidentical donors.¹⁵ In this study, we outlined the dynamic of intestinal microbiota over allo-HCT and explored its association with clinical outcomes after allo-HCT in a large Chinese cohort.

MATERIALS AND METHODS

Study Design and Sample Collection

Patients who underwent allo-HCT at the Department of Hematology, the first medical center of Chinese PLA General Hospital, between November 2018 and July 2020 were prospectively enrolled. Written informed consent was obtained from all patients for the use of their stool samples and clinical data for research before specimen collection. This study was reviewed and approved by the Medical Ethics Review Board of Chinese PLA General Hospital. The general design and principles of sample collection are provided in supporting information.

Transplantation Procedures

Nearly all patients received unmodified peripheral blood stem cells grafts that were mobilized with recombinant human granulocyte colony-stimulating factor as the sole source of graft, with the exception for a few patients with aplastic anemia who also received graft from granulocyte colony-stimulating factor-primed bone marrow. Three myeloablative conditioning regimens were used in this study: modified busulfan/cyclophosphamide (CY), total

body irradiation/CY, and modified fludarabine (Flu)/CY, as previous described.¹⁶ All patients received ATG-based GVHD prophylaxis (thymoglobulin, rabbit; Genzyme Europe BV, Naarden, the Netherlands). In detail, the dosage of ATG was 2.5 mg/kg/d on days -5 to -2 for a total dosage of 10 mg/kg for patients using haploidentical (haplo-HCT) or unrelated donor (MUD-HCT). For patients using human leukocyte antigen-matched sibling donor HCT, the total dosage of ATG was 5 mg/kg. The use of cyclosporine, mycophenolate, and short-term methotrexate was as indicated in our previous study.¹⁷ Detailed information about antibiotic use is provided in the supporting information.

Sample Processing and Analyzing

Total bacterial genomic DNA was extracted from each fecal specimen. The V3 to V4 region of the 16S rRNA gene was amplified by polymerase chain reaction using modified universal bacterial primers. Then polymerase chain reaction products were purified and sequenced on the Illumina MiSeq platform (Illumina, Inc, San Diego, California).¹⁸ Sequence data were filtered for quality and grouped into operational taxonomic units with the taxonomic threshold of 97% similarity by using the UCLUST algorithm.¹⁹ The Shannon diversity index (SI), an ecological estimate of α diversity that integrates both the information on the richness and the uniformity of an ecosystem, was used to estimate the microbial diversity. Intestinal domination was defined as a certain bacterial genus accounting for >30% of sequences and was the most abundant genus.

Statistical Analysis

Trends on the microbial diversity over different time points of allo-HCT were described with moving average filtering. Curvewise 95% confidence intervals were also calculated. Descriptive analysis was used to compare baseline characteristics of patients in different diversity groups. Patients' disease risk was calculated using the Disease Risk Index system.²⁰ χ^2 tests were performed for categorical variables, and Mann-Whitney U tests were performed for continuous variables. Analysis of clinical outcomes began at the time of stem cell infusion (day 0) and continued up to 3 years of allo-HCT or last follow-up for survivors. Overall survival (OS) and progression-free survival (PFS) were performed with the Kaplan-Meier method and compared using the log-rank test. For transplantation-related mortality (TRM), relapse, and GVHD, their cumulative incidences were estimated with competing-risk analysis and compared with a Gray's test. Cox proportional

hazards analysis was used to determine any significant associations between baseline variables and the outcomes of interest. Variables with $P < .20$ were entered in Cox multivariate analysis, and only those with $P \leq .05$ were retained in the final model. All analyses were performed with R software (cmprsk and survival package).

RESULTS

Dynamics of Gut Bacterial Diversity During Allo-HCT

During our enrollment period, a total of 623 stool samples were prospectively collected from 101 patients. After exclusion of unqualified samples and samples from 1 patient who died before engraftment, 603 samples from 100 patients were eligible for further analysis. The median samples per patient number was 5. The clinical characteristics of these patients are summarized in Supporting Table 1. There was a wide range of individual variation in the bacterial diversity reflected by the Shannon index values. Nevertheless, the overall average SI began to decline as early as the start of the conditioning regimen (Fig. 1A). It decreased over the course of allo-HCT and decreased to the lowest point on day 12 after transplantation. Then the bacterial diversity gradually increased over time. A similar pattern in the shift of the median diversity among the 4 time periods during allo-HCT was also observed (Fig. 1B). Despite the great variation in SI values, more patients were found to develop lower gut diversity in the engraftment period.

Correlation Between Clinical Outcomes and Intestinal Microbiota Diversity During the Engraftment Period

We focused on the 86 patients who produced 314 samples during the engraftment period. For patients with multiple samples in this period, the median value (SI) of each patient was used. Based on the median value (SI = 2.61) of all patients' microbial diversity, patients were divided into a high-diversity group (SI > 2.61) and a low-diversity group (SI < 2.61). Clinical characteristics of these 86 patients are shown in Table 1. There were no significant differences for most variables. Nearly all patients received fluoroquinolone and β -lactams. More than 60% of patients received vancomycin. No differences were found about the use of these 3 kinds of antibiotics between the high- and low-diversity groups. Nevertheless, patients receiving matched sibling donor HCT were more likely to have high bacterial diversity ($P = .024$).

After a median follow-up of 20 months, 7 patients died in the high-diversity group, whereas there were 16 deaths in the low-diversity group. The estimated 2-year OS in the high-diversity group was higher than that in the low-diversity group (83.7% vs 60.6%, $P = .026$; Fig. 2A). Although there was no significant difference, the 2-year PFS of patients with relatively high diversity seemed to be higher than that in the low-diversity group ($P = .072$; Fig. 2B). Cox proportional hazards regression analysis for OS is shown in Table 2. After adjusting for disease risk, pretransplant comorbidity, and previous chemotherapy before HCT, low microbial diversity (hazard ratio [HR],

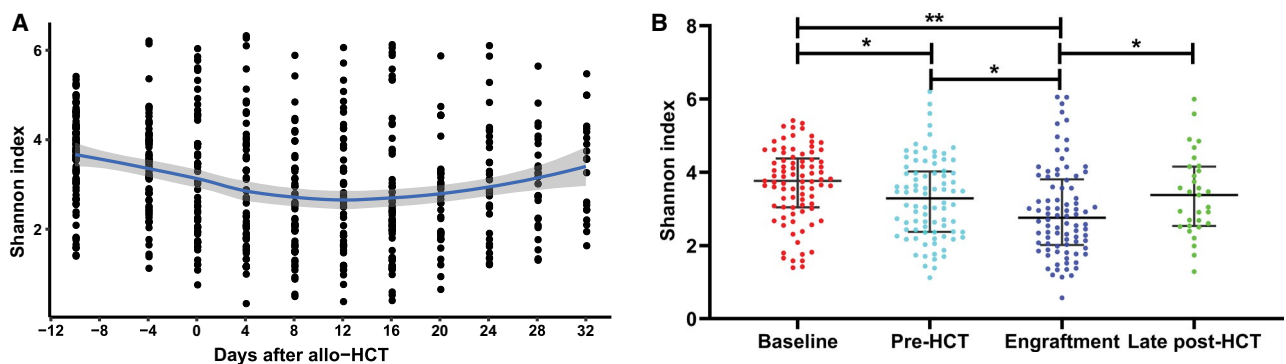


Figure 1. Alterations of intestinal microbiota diversity in allo-HCT recipients. (A) Gut microbial diversity was calculated with the Shannon diversity index for each fecal specimen of each patient and plotted over day of allo-HCT. The trend was plotted using moving average filtering with 95% confidence intervals. (B) The median Shannon index of samples collected among the 4 time periods relative to allo-HCT. Each period column is represented by its median value and interquartile range. There was only 1 sample collected for each patient at the baseline, whereas, for patients with multiple samples among the other 3 time periods relative to allo-HCT, the median value (Shannon diversity index) of each patient was used. Statistical analyses were performed using nonparametric Wilcoxon Mann-Whitney test. * $P < .05$; ** $P < .01$; allo-HCT indicates allogeneic hematopoietic stem cell transplantation; HCT indicates hematopoietic stem cell transplantation.

TABLE 1. Clinical Characteristics of 86 Patients Who Received Allo-HCT

Characteristics	High Diversity (N = 43)	Low Diversity (N = 43)	P
Age, median (range), y	31 (13-60)	34 (12-62)	.462
Gender (female), No. (%)	10 (23.3)	6 (14.0)	.268
Underlying disease, No. (%)			
AML	19 (44.2)	21 (48.8)	.658
MDS	1 (2.3)	3 (7.0)	
ALL	19 (44.2)	15 (34.9)	
MPN or MDS/MPN	2 (4.7)	2 (4.7)	
Aplastic anemia	1 (2.3)	1 (2.3)	
Non-Hodgkin lymphoma	1 (2.3)	1 (2.3)	
Disease Risk Index, No. (%)			.267
Low	24 (55.8)	17 (39.5)	
Intermediate	16 (37.2)	20 (46.5)	
High	3 (7.0)	6 (14.0)	
Pretransplant comorbidity (HCT-CI), No. (%)			.610
0	16 (37.2)	18 (41.9)	
1-2	24 (55.8)	20 (46.5)	
≥3	3 (7.0)	5 (11.6)	
Disease status at HCT, No. (%)			.366
Complete remission	38 (88.4)	35 (81.4)	
Noncomplete remission	5 (11.6)	8 (18.6)	
No. of cycles of chemotherapy before HCT, No. (%)			.167
≤4	32 (74.4)	26 (60.5)	
>4	11 (25.6)	17 (39.5)	
Previous chemotherapy, No. (%)			1.0
Yes	40 (93.0)	40 (93.0)	
No	3 (7.0)	3 (7.0)	
Median time from diagnosis to HCT (range), mo	5 (1-19)	6 (1-29)	.429
Donor-recipient ABO match, No. (%)			.829
Match	23 (53.5)	22 (51.2)	
Mismatch	20 (46.5)	21 (48.8)	
Allo-HCT type, No. (%)			.024
MSD-HCT	10 (23.3)	2 (4.7)	
Haplo-HCT	32 (74.4)	37 (86.0)	
MUD-HCT	1 (2.3)	4 (9.3)	
Conditioning regimens, No. (%)			.413
TBI/Cy based	7 (16.3)	6 (14.0)	
BuCy based	35 (81.4)	36 (83.7)	
Flu/Cy	1 (2.3)	1 (2.3)	
Days to engraftment, median (range), d			
Neutrophil engraftment	14 (9-20)	13 (7-23)	.764
Platelet engraftment	13 (8-39)	13 (8-39)	.701
Antibiotic exposure ^a			
Vancomycin	28 (65.1)	26 (60.5)	.655
Fluoroquinolone	42 (97.7)	41 (95.3)	.557
β-lactam ^b	43 (100.0)	43 (100.0)	NA
Follow-up of survivors, mo			.578
Median	20	20	
Range	12-33	12-31	

Abbreviations: ALL, acute lymphoid leukemia; allo-HCT, allogeneic hematopoietic stem cell transplantation; AML, acute myeloid leukemia; ATG, antithymocyte globulin; Bu/Cy, busulfan/cyclophosphamide; FB/Cy, fludarabine/cyclophosphamide; haplo-HCT, allogeneic hematopoietic stem cell transplantation using human leukocyte antigen-haploidentical donor; HCT-CI, hematopoietic cell transplantation comorbidity index; HLA, human leukocyte antigen; MDS, myelodysplastic syndrome; MPN, myeloproliferative neoplasm; MSD-HCT, allogeneic hematopoietic stem cell transplantation using human leukocyte antigen-matched sibling donor; MUD-HCT, allogeneic hematopoietic stem cell transplantation using human leukocyte antigen-matched unrelated donor; NA, not applicable; TBI/Cy, total body irradiation/cyclophosphamide.

^aAssessed from the start of the conditioning regimen to the last specimen collected for each patient.

^bβ-lactams include cephalosporins, β-lactam-β-lactamase combinations, and carbapenems.

2.62; 95% confidence interval [CI], 1.06-6.49) during the engraftment period and noncomplete remission before HCT (HR, 2.81; 95% CI, 1.13-6.99) were independent predictors of all-cause death in the multivariate analysis.

Further analysis showed that the estimated 2-year TRM in the low-diversity group was higher than that in the low-diversity group (20.0% vs 4.7%, $P = .04$; Fig. 2C). Low microbial diversity was an independent risk factor for TRM both in univariable and multivariate analysis (Table 3). Conversely, there was no significant difference in relapse rate (18.9% in the high-diversity group vs 20.9% in the low-diversity group, $P = .651$; Fig. 2D). No differences were found in the incidence of II-IV and III-IV acute GVHD (aGVHD) between these 2 groups (Supporting Fig. 1). We also further explored the possible influence of intestinal microbiota diversity on gut aGVHD and chronic GVHD. However, no significant difference was found (Supporting Figs. 2 and 3).

The causes of death of all patients were also analyzed (Supporting Table 2). More patients (6 of 43) were found to die of infections in the low-diversity group, whereas only 1 patient died because of infection in the high-diversity group. On the contrary, incidences of death because of disease relapse and aGVHD were more evenly distributed. The incidence of bacterial, viral, or fungal infections after allo-HCT was also studied (Supporting Table 3). Nevertheless, no differences were found. Taken together, our results indicate that worse outcomes of patients with low bacterial gut diversity during the engraftment period were more likely to be associated with severe infection.

Alterations to the Composition of Fecal Microbiota Over Allo-HCT

The prevalence of intestinal domination by a certain bacterial genus was about 80% of all patients before the start of conditioning (Fig. 3A). As the pattern of the Shannon diversity, the incidence of intestinal domination increased to approximately 95% of all patients during the engraftment period and then began to decrease. The beneficial genus *Bacteroides* accounted for the majority of patients' intestinal bacteria before conditioning and decreased over the course of allo-HCT (Fig. 3B,C). Conversely, the abundance of pathogenic genera including *Enterococcus*, *Klebsiella*, and *Escherichia* was found to be increased during the engraftment period. These data indicated that the compositional change over all-HCT was characterized by the decreased abundance of beneficial microbiota and the increased abundance of pathogenic genera.

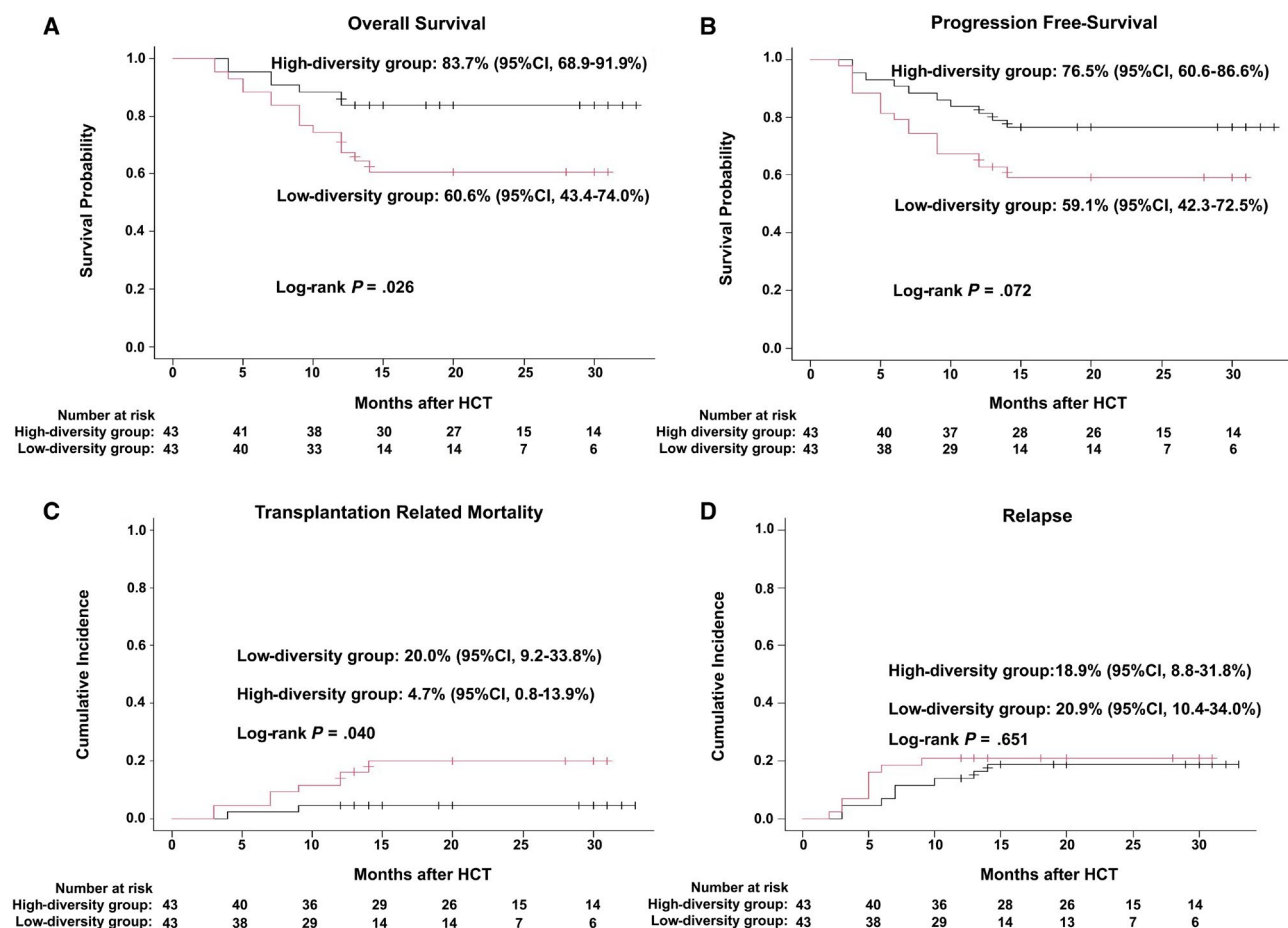


Figure 2. Kaplan-Meier analysis of patients with different intestinal microbiota diversity. (A) Overall survival, (B) progression-free survival, (C) transplantation-related mortality, and (D) relapse. HCT indicates, allogeneic hematopoietic stem cell transplantation.

DISCUSSION

In this prospective, observational, single-center study, we outlined the dynamics of intestinal microbiota of patients in ATG-based myeloablative allo-HCT. Our study indicated that the gut dysbiosis over allo-HCT also featured disruption of bacterial diversity in our transplantation platform. Low-gut microbial diversity during the engraftment period was found to be an independent predictor of poor survival. The poor survival of patients with low bacterial diversity was demonstrated to be associated with TRM, but not with disease relapse or progression. Crucially, further analysis about the causes of death of patients in our cohort suggested that poor outcome of patients with low diversity was more likely because of infection but not GVHD. Compositional analysis on fecal microbiota revealed the overall shift pattern from beneficial bacteria to pathogenic genera over the course of allo-HCT, helping to explain the association between the intestinal flora and clinical outcomes. Despite the overall

similar pattern that was reported in single-center and multicenter studies,^{5,21-23} our study featured the alteration of intestinal microbiota over allo-HCT and its association with clinical outcomes in platforms of ATG-based myeloablative allo-HCT.

During the course of allo-HCT, the intestinal microenvironment was severely disrupted because of the toxicities of the conditioning regimens that involving various intensity of chemotherapy and/or total body irradiation, and the administration of all kinds of antibiotics for antibacterial prophylaxis and the treatment of febrile neutropenia. This kind of gut dysbiosis often deteriorated because of the further damage on the intestinal epithelial cells by the alloreactive T lymphocytes in the grafts.²² Therefore, the number of T cells in the grafts, or in other words, the way of T-cell-depleting methods, was also an important determinant for gut dysbiosis. Considering the great differences in the conditioning regimen, antibiotic use, and T-cell

TABLE 2. Predictors of Overall Survival

Characteristic	Univariate		Multivariate	
	Hazard Ratio	P	Hazard Ratio	P
Age (y >40)	1.99 (0.88-4.52)	.099	2.025 (0.88-4.66)	.097
Sex (female)	0.59 (0.18-2.01)	.403		
Underlying disease				
AML/MDS	1.0	—	1.0	—
ALL	0.50 (0.19-1.31)	.161	0.63 (0.22-1.79)	.383
Other	1.15 (0.34-4.01)	.825	0.17 (0.02-1.30)	.087
Disease risk ^a				
Low	1.0	—	1.0	—
Intermediate	2.83 (1.07-7.44)	.035	1.60 (0.55-4.71)	.391
High	4.02 (1.13-14.28)	.032	0.60 (0.08-4.61)	.621
Pretransplant comorbidity (HCT-CI ≥2)	3.03 (1.24-7.39)	.014	2.24 (0.90-5.56)	.081
Disease status at HCT (non-CR)	4.24 (1.79-10.3)	.001	2.81 (1.13-6.99)	.026
No. of cycles of chemotherapy before HCT (>4)	0.71 (0.28-1.81)	.474		
Previous chemotherapy (yes)	0.31 (0.11-0.92)	.035	1.32 (0.29-6.04)	.725
Months from diagnosis to HCT (>5)	1.38 (0.61-3.14)	.437		
Donor-recipient ABO match				
Match	1.0	—		
Mismatch	1.18 (0.52-2.68)	.691		
Allo-HCT type (MSD-HCT vs other)	1.90 (0.45-8.12)	.384		
Conditioning regimens				
BuCy based	1.0	—		
TBI/Cy based	0.79 (0.23-2.66)	.700		
ATG dosage				
Low	1.0	—		
High	1.90 (0.45-8.12)	.384		
Antibiotic exposure ^a				
Vancomycin	0.547 (0.216-1.39)	.204		
Fluoroquinolone	2.30 (0.538-9.53)	.261		
β-lactam ^b	NA			
Microbial diversity (engraftment)				
High diversity	1.0	—	1.0	—
Low diversity	2.61 (1.07-6.35)	.035	2.62 (1.06-6.49)	.038

Abbreviations: ALL, acute lymphoid leukemia; allo-HCT, allogeneic hematopoietic stem cell transplantation; AML, acute myeloid leukemia; ATG, antithymocyte globulin; Bu/Cy, busulfan/cyclophosphamide; CR, complete remission; HCT-CI, hematopoietic-cell transplantation comorbidity index; HLA, human leukocyte antigen; MDS, myelodysplastic syndrome; MSD-HCT, allo-HCT using human leukocyte antigen-matched sibling donor; TBI/Cy, total body irradiation/cyclophosphamide.

^aAssessed from the start of the conditioning regimen to the last specimen collected for each patient.

^bβ-lactams include cephalosporins, β-lactam-β-lactamase combinations, and carbapenems.

depleting methods, our study featured the platforms of ATG-based myeloablative allo-HCT. The dynamics of the intestinal flora over allo-HCT in our platform was generally similar to previous reports on platforms involving nonmyeloablative conditioning and mainly T-cell depletion or postcyclophosphamide-based T-cell-repleted GVHD prophylaxis regimens.^{5,21,23} Therefore, our study confirmed and extended these observations.

Previous studies pointed out that the poor outcome of patients with lower microbiota diversity was more prominently correlated with TRM and GVHD-related death for patients receiving T-cell-repleted grafts.⁵ Other studies using T-cell-repleted grafts also demonstrated that low diversity of the gut microbiota was a predictor of aGVHD-related death.^{9,24,25} These results indicated that this association may depend on the presence of alloreactive T lymphocytes.⁵ Our results suggested that the poor outcome of patients with low diversity was more likely

because of infection, but not aGVHD. Two reasons may help to explain this difference. First, the correlation between the alteration of gut microbiota and infection has been reported.^{8,26,27} The conditioning regimen in our study was all myeloablative. So, the intensity was higher than that in most of the above studies. Second, GVHD prophylaxis in our study was an ATG-based in vivo T-cell-depleting method. The dose of ATG was relatively higher. The damage of alloreactive lymphocytes on intestinal epithelial cells was probably reduced to a lesser extent. However, the association between diversity and clinical outcomes was not found in subgroup analysis on platforms using T-cell-depleted grafts. T-cell-depleted allo-HCT using CD34-selected grafts was also performed with a myeloablative conditioning regimen. Although further checking the detailed drug use within this conditioning regimen of T-cell-depleted allo-HCT platforms was performed, the cytotoxic effects of our conditioning

TABLE 3. Predictors of Transplantation-Related Mortality

Characteristic	Univariate		Multivariate	
	Hazard Ratio	P	Hazard Ratio	P
Age (y >40)	2.25 (0.65-7.80)	.201		
Sex (female)	0.42 (0.05-3.29)	.405		
Underlying disease				
AML/MDS	1.0			
ALL	0.70 (0.17-2.93)	.624		
Other	2.13 (0.41-11.00)	.365		
Disease risk ^a				
Low	1.0			
Intermediate	1.76 (0.47-6.55)	.402		
High	1.65 (0.18-14.81)	.656		
Pretransplant comorbidity (HCT-CI ≥2)	3.54 (0.91-13.81)	.068	2.69 (0.64-11.39)	.179
Disease status at HCT (non-CR)	3.82 (0.97-14.93)	.054	1.89 (0.15-23.86)	.621
No. of cycles of chemotherapy before HCT (>4)	0.82 (0.21-3.16)	.768		
Previous chemotherapy (yes)	0.25 (0.05-1.16)	.076	1.32 (0.29-6.04)	.725
Months from diagnosis to HCT (>5)	2.12 (0.60-7.53)	.244		
Donor-recipient ABO match				
Match	1.0			
Mismatch	1.12 (0.32-3.88)	.855		
Allo-HCT type (MSD-HCT vs other)	5.67 (0.27-15.20)	.345		
Conditioning regimens				
BuCy based	1.0			
TBI/Cy based	2.28 (0.57-9.13)	.245		
ATG dosage				
Low	1.0			
High	5.67 (0.27-15.20)	.345		
Antibiotic exposure ^a				
Vancomycin	0.376 (0.080-1.77)			
Fluoroquinolone	2.55 (0.324-2.2)	.375		
β-lactam ^b	NA			
Microbial diversity (engraftment)				
High diversity	1.0			
Low diversity	4.84 (1.02-22.86)	.047	4.95 (1.03-23.76)	.046

Abbreviations: ALL, acute lymphoid leukemia; allo-HCT, allogeneic hematopoietic stem cell transplantation; AML, acute myeloid leukemia; ATG, antithymocyte globulin; Bu/Cy, busulfan/cyclophosphamide; CR, complete remission; HCT-CI, hematopoietic cell transplantation comorbidity index; HLA, human leukocyte antigen; MDS, myelodysplastic syndrome; MSD-HCT, allo-HCT using human leukocyte antigen-matched sibling donor; TBI/Cy, total body irradiation/cyclophosphamide.

^aAssessed from the start of the conditioning regimen to the last specimen collected for each patient.

^bβ-lactams include cephalosporins, β-lactam-β-lactamase combinations, and carbapenems.

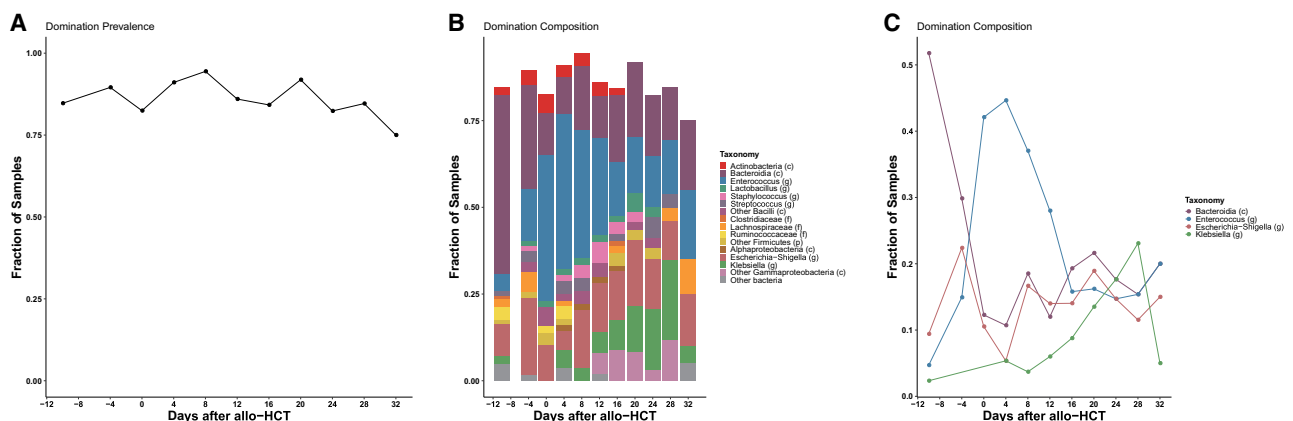


Figure 3. Compositional change of fecal microbiota over allo-HCT. (A) The prevalence of intestinal domination by a certain bacterial genus over time of allo-HCT. Intestinal domination was defined as a certain bacterial genus accounting for >30% of sequences and was the most abundant genus. (B) The bacterial composition of the dominant genera of samples collected over the course of allo-HCT. (C) The fraction of the most abundant of microbiota among samples over allo-HCT. allo-HCT indicates allogeneic hematopoietic stem cell transplantation.

regimen were found to be higher. This may partially explain why this association was not found in platforms using T-cell-depleted grafts. However, all these questions need to be answered in a prospective randomized control study.

Another important determinant that greatly influenced gut flora was antibiotic use during the course of allo-HCT. Early studies suggested that gut decontamination with antibiotics helps to reduce the incidence of GVHD and infection.^{28,29} On the contrary, recent evidence indicated that this procedure may be detrimental.³⁰ This is consistent with current paradigm that lower bacterial diversity was associated with poor outcomes. Recent studies also implied that the use of broad-spectrum antibiotics dramatically disrupts the intestinal flora and was also a contributor for poor outcome.^{21,31} However, the associations between antibiotic use and clinical outcomes were not found in our study. This is consistent with a previous study.²² Exposure to piperacillin-tazobactam or meropenem was also found to be associated with a decrease in diversity during allo-HCT.⁵ Nevertheless, no correlation was found between 1 certain antibiotic use and intestinal bacterial diversity in our study. Several factors may help to explain these differences. First, the basic principle for antibiotic use was not consistent among different transplantation centers. Second, the purpose of antibiotic use for each patient was different. It could be targeted or empiric. In our study, nearly all patients experienced neutropenia fever. All patients received β -lactams. Despite the detrimental effect of antibiotics on intestinal flora, it is still very difficult to reduce the use of them during allo-HCT.

We further examined the compositional change of fecal microbiota. Consistent with previous studies,^{5,23} our transplantation platform also generally featured the decreased abundance of beneficial microbiota and the increased abundance of pathogenic genera. Nevertheless, the prevalence of intestinal domination and the percentage of *Bacteroides* domination in our study were higher than that in the Memorial Sloan Kettering Cancer Center cohort. Considering lots of factors affecting intestinal flora, these differences might be related to different dietary habits in different countries. This kind of compositional shift over allo-HCT can partially explain why some patients were more likely to die of infection. The prognostic value of each bacterial genus such as enterococcus was also investigated.³² To improve the prognostic values for clinical outcomes, studies also tried to establish microbiota-compositional score systems.^{5,33} Nevertheless,

it is currently unrealistic to predict clinical outcome through precise characterization of an individual intestinal microbiota.

Limitations of our study included the difference of transplantation using 3 types of donors, a low number of patients from only a single center, and fewer data about antibiotic use under various conditions. The advantages included the large sample size at different time points over allo-HCT, universal myeloablative conditioning, and ATG-based GVHD prophylaxis regimens. In summary, our study characterizes the alteration of intestinal microbiota over allo-HCT and points out that low intestinal diversity was also associated with a poor outcome on platforms with myeloablative conditioning and ATG-based GVHD prophylaxis regimens. Our study further stresses the rational interventions to restore the microbial equilibrium. Unmanipulated fecal microbiota transplantation had been reported to be effective in selective patients. Administration of both probiotics and prebiotics also proved to be useful.³⁴ Nevertheless, these strategies still need to be evaluated in large prospective controlled studies.

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CONFLICT OF INTEREST DISCLOSURES

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AUTHOR CONTRIBUTIONS

Daihong Liu: Study concept, supervision, and design; data collection; statistical analysis; manuscript writing; and funding acquisition. **Zhenyang Gu:** Data collection, statistical analysis, analysis, data interpretation, and manuscript writing. **Qian Xiong:** Data collection and analysis and data interpretation. **Lu Wang:** Data collection and analysis and data interpretation. **Lili Wang:** Data collection and analysis and data interpretation. **Fei Li:** Data collection and analysis and data interpretation. **Cheng Hou:** Data collection and analysis and data interpretation. **Baoli Zhu:** Study concept and design, analysis and data interpretation, and methodology. **Liping Dou:** Study concept and design, analysis and data interpretation, and methodology.

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